

Midterm Study Notes

Section 2.2 & 2.3 Topics

1. Definition of $a \mid b$.

For integers a and b , $a \mid b$ if there exists an integer k such that $b = ak$. Example: $4 \mid 20$ because $20 = 4 \cdot 5$, but $4 \nmid 18$.

2. Simple Properties of Division.

Using $a \mid b \iff b = ak$:

- *Transitivity*: If $a \mid b$ and $b \mid c$, then $a \mid c$.
- *Additivity*: If $a \mid b$ and $a \mid c$, then $a \mid (b + c)$.
- *Cancellation*: If $a \mid b$ and $a \mid c$, then $a \mid (mb + nc)$ for any integers m, n .

3. Division Algorithm.

For a and $b \neq 0$, there exist unique q, r with $0 \leq r < |b|$ such that

$$a = bq + r.$$

Example: $23 = 5 \cdot 4 + 3$.

4. Euclidean Algorithm (Finding $\gcd(a, b)$).

Repeatedly apply the Division Algorithm:

$$\gcd(a, b) = \gcd(b, r),$$

until remainder = 0. Example: $\gcd(48, 18)$:

$$48 = 18 \cdot 2 + 12, \quad 18 = 12 \cdot 1 + 6, \quad 12 = 6 \cdot 2 + 0 \implies \gcd(48, 18) = 6.$$

5. Extended Euclidean Algorithm (Solving $ax + by = \gcd(a, b)$).

Keep track of remainders to express $\gcd(a, b)$ as $ax + by$. Example: $\gcd(48, 18) = 6$.

$$48 = 18 \cdot 2 + 12 \implies 12 = 48 - 2 \cdot 18,$$

$$18 = 12 \cdot 1 + 6 \implies 6 = 18 - 12 = 18 - (48 - 2 \cdot 18) = 3 \cdot 18 - 48.$$

Hence $6 = (-1) \cdot 48 + 3 \cdot 18$, so $x = -1, y = 3$.

6. Finding All Divisors of an Integer.

Example: For 36, the positive divisors are $\{1, 2, 3, 4, 6, 9, 12, 18, 36\}$. Negatives are their negatives.

7. Fundamental Theorem of Arithmetic (Prime Factorization).

Every positive integer has a unique prime factorization. Example: $84 = 2^2 \times 3 \times 7$.

8. Definition of $a \equiv b \pmod{m}$.

$a \equiv b \pmod{m}$ if $m \mid (a - b)$. Equivalently, a and b have the same remainder mod m . Example: $14 \equiv 2 \pmod{4}$.

9. Simple Properties of Congruences.

If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then:

- $a + c \equiv b + d \pmod{m}$.
- $ac \equiv bd \pmod{m}$.
- $-a \equiv -b \pmod{m}$.

10. Modular Arithmetic in $\mathbb{Z}/m\mathbb{Z}$.

$\mathbb{Z}/m\mathbb{Z} = \{[0], [1], \dots, [m-1]\}$. Operations:

$$[a] + [b] = [a + b], \quad -[a] = [-a], \quad [a] \cdot [b] = [ab].$$

Example: In $\mathbb{Z}/5\mathbb{Z}$, $[2] + [4] = [6] = [1]$.

11. Inverses in $\mathbb{Z}/m\mathbb{Z}$ Using Extended Euclidean Algorithm.

If $\gcd(a, m) = 1$, then $[a]$ has a multiplicative inverse in $\mathbb{Z}/m\mathbb{Z}$. Example: In $\mathbb{Z}/7\mathbb{Z}$, to find $[3]^{-1}$, solve $3x + 7y = 1$. One solution is $x = 5$, so $[3]^{-1} = [5]$.

Section 3.1 Topics

1. Definition of $\text{Sym}(X)$.

$\text{Sym}(X)$ is the set of all bijective functions $f : X \rightarrow X$. It forms a group under composition. Example: if $X = \{1, 2, 3\}$, any permutation (rearrangement) is in $\text{Sym}(X)$.

2. Notations for Permutations in S_n .

Two-row notation:

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & \dots & n \\ \sigma(1) & \sigma(2) & \sigma(3) & \dots & \sigma(n) \end{pmatrix}.$$

Cycle notation: $(a_1 a_2 \dots a_r)$ means $\sigma(a_1) = a_2, \sigma(a_2) = a_3, \dots, \sigma(a_r) = a_1$. Example:

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} = (1\ 3\ 4\ 2).$$

3. Composing Two Permutations.

$\tau \circ \sigma$ means apply σ first, then τ . Example:

$$\sigma = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}, \quad \tau = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}.$$

Then $\tau(\sigma(1)) = \tau(2) = 1$, $\tau(\sigma(2)) = \tau(3) = 2$, $\tau(\sigma(3)) = \tau(1) = 3$. So $\tau \circ \sigma = (1\ 2\ 3)$.

4. More Complicated Permutation Compositions.

Cycle Inverse: $(a_1 a_2 \dots a_r)^{-1} = (a_r a_{r-1} \dots a_1)$.

Conjugation: $\tau(a_1 a_2 \dots a_r)\tau^{-1} = (\tau(a_1) \tau(a_2) \dots \tau(a_r))$. Example: $(1\ 2\ 3)^{-1} = (3\ 2\ 1)$.

Section 3.2, 3.3, 4.1 Topics

1. Axioms of Semigroups, Monoids, and Groups.

- *Semigroup:* A set S with an associative binary operation.
- *Monoid:* A semigroup with identity e where $e \cdot a = a \cdot e = a$.
- *Group:* A monoid in which every element has an inverse.

2. Proving Simple Properties from Axioms.

Example: from associativity and inverses, we get *cancellation*: if $ab = ac$, then $b = c$. In $(\mathbb{Z}, +)$, identity is 0, inverse of a is $-a$.

4. The Dihedral Group D_n .

Group of symmetries of a regular n -gon:

$$D_n = \{1, r, r^2, \dots, r^{n-1}, s, sr, sr^2, \dots, sr^{n-1}\}, \quad r^n = 1, s^2 = 1, srs = r^{-1}.$$

Interpretation: r is rotation by $360^\circ/n$; s a reflection. Example: D_4 (square) has 8 elements: $\{1, r, r^2, r^3, s, sr, sr^2, sr^3\}$.

5. Definition of a Subgroup.

$H \subseteq G$ is a subgroup if:

- $H \neq \emptyset$,
- If $a, b \in H$, then $ab \in H$,
- If $a \in H$, then $a^{-1} \in H$.

6. Using the Subgroup Criterion to Find All Subgroups.

Example: In $\mathbb{Z}_{12}$ under addition, subgroups are $\langle d \rangle$ for each d dividing 12. E.g. $\langle 4 \rangle = \{0, 4, 8\}$.

7. Order of an Element.

The order of a is the smallest positive integer n such that $a^n = e$. Also the size of the cyclic subgroup generated by a . Example: In $(\mathbb{Z}/12\mathbb{Z}, +)$, $[3]$ has order 4 because $3 \cdot 4 = 12 \equiv 0 \pmod{12}$.

8. Left Coset aH .

If H is a subgroup of G and $a \in G$, then $aH = \{ah : h \in H\}$. Example: In $\mathbb{Z}_{12}$, $H = \{0, 4, 8\}$, then $[3] + H = \{[3], [7], [11]\}$.

9. Lagrange's Theorem.

If H is a finite subgroup of G , $|H|$ divides $|G|$. Thus if $|G| = 12$, subgroups can only have sizes dividing 12: 1, 2, 3, 4, 6, 12.